Type Testing of Rising Stem Valves Equipped with Flexible Graphite Packing for Fugitive Emissions

API STANDARD 624
1ST EDITION, APRIL 2012

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INTRODUCTION

The purpose of this API standard is to establish a uniform procedure for the evaluation of emission performance of process valves. The testing program will provide a basis for the comparison of the emissions and performance of process valves.

Use of this standard assumes the execution of its provisions is entrusted to appropriately qualified and experienced personnel because it calls for procedures that can be injurious to health if adequate precautions are not taken. This standard refers only to technical suitability and does not absolve the user from legal obligations relating to health and safety at any stage of the procedure.

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1. Scope

This API standard specifies the requirements and acceptance criteria (100 ppmv) for fugitive emission type testing of rising and rising-rotating stem valves equipped with packing previously tested in accordance with API Standard 622. Packing shall be suitable for use at service temperatures –29°C to 538°C (–20°F to 1000°F). The type testing requirements contained herein are based upon elements of EPA Method 21.

2. Normative References

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

API Standard 598, *Valve Inspection and Testing*

API Standard 600, *Steel Gate Valves—Flanged and Butt-welding Ends, Bolted Bonnets*

API Standard 602, *Steel Gate, Globe and Check Valves for Sizes DN 100 and Smaller for the Petroleum and Natural Gas Industries*

API Standard 622, *Type Testing of Process Valve Packing for Fugitive Emissions*

ASME B16.34, *Valves—Flanged, Threaded, and Welding End*

EPA Method 21, *Determination of Volatile Organic Compound Leaks*

3. Terms and Definitions

For the purposes of this standard, the following definitions shall apply.

3.1. **Ambient temperature**
Temperature that is between 15°C to 40°C (60°F to 104°F)

3.2. **Auxiliary connection**
Drain(s), vent(s) or sealant injection port(s)

3.3. **Bolting torque**
The amount of twisting or turning effort (expressed as N-m, ft-lb, or in-lb) required to tighten a threaded fastener.

3.4. **Dynamic leak measurement**
Measurement of leakage taken while the stem is travelling through an opening and closing cycle.

3.5. **EPA Method 21**
A leak check method established by the United States Environmental Protection Agency (EPA) for performing emissions measurements on equipment such as valves, pumps, and flanges.

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3.6. **Emissions**
Gaseous leak given off by a piece of equipment used in reference to volatile organic compounds (VOCs) and expressed in parts per million volumetric (ppmv or ppm) for methane.

3.7. **Leakage**
Measurable amount of test medium escaping from the valve.

3.8. **Manufacturing facility**
Location of final assembly, inspection, and testing of the valve selected for evaluation.

3.9. **Mechanical cycle**
A motion of the stem simulating the movement of a valve obturator from the fully closed position to the fully open position and returning to the fully closed position.

3.10. **Obturator**
Part of a valve, such as a wedge, disc, or plug, which is positioned in the flow stream to permit or prevent flow.

3.11. Purchaser
A person, group, company, agency, corporation, or designated representative responsible for valve acceptance.

3.12. **Rising stem valve**
Valve in which the movement of the stem is in an axial direction with no rotation.

3.13. **Rising-rotating stem valve**
Valve in which the movement of the stem is in an axial direction with rotation.

3.14. Running torque
The amount of torque required to move a valve stem through the packing.

3.15. Static leak measurement
Measurement of leakage taken when the stem is not moving.

3.16. **Stem**
Metal rod that connects the obturator of a valve to a handwheel, handle, or actuator.

3.17. **Thermal cycle**
A change in temperature from ambient temperature, to elevated temperature, and back to ambient temperature.

3.18. **Type testing**
A test conducted to assess the performance of a specific valve design at prescribed temperatures and pressures.

4. Valve Selection & Test Preparation

4.1. Test valve shall be randomly selected from manufacturer or distributor stock where such stock is available. For valves not in stock, the manufacturer shall certify that the test valve was not modified in any way to meet type test requirements and is a typical representation of the manufacturer's stock product. Valve selection shall be approved by the purchaser.

4.2. An API 598 certificate shall be provided with the test valve.

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4.3. Body-bonnet and gland bolting shall be tightened in accordance with manufacturer's specifications. Prior to type testing, it is permissible to operate the test valve multiple times (at temperature and under pressure) to ensure that the packing is properly installed and consolidated in the stuffing box. (Note: The purpose of this operation is to minimize the likelihood that the packing would require adjustment during type testing to meet the permissible leakage limit.)

4.3.1. As a safety precaution, the air in the valve cavity shall be purged with an inert gas prior to starting the testing.

4.3.2. The preferred stem orientation for a test valve is horizontal.

4.3.3. The test medium used shall be methane 97% minimum purity. **Note: Methane used during testing is a pressurized flammable gas and appropriate safety measures shall be taken.**

5. Type Testing

5.1. Valves shall be subjected to a total of 310 mechanical cycles and 3 thermal cycles per Figure 1. The valve stroke shall be approximately 80% of the mechanical cycle. Mechanical cycling shall begin with the valve at ambient temperature. An optional low temperature test at -29°C (-20°F) may be performed if requested by the purchaser.

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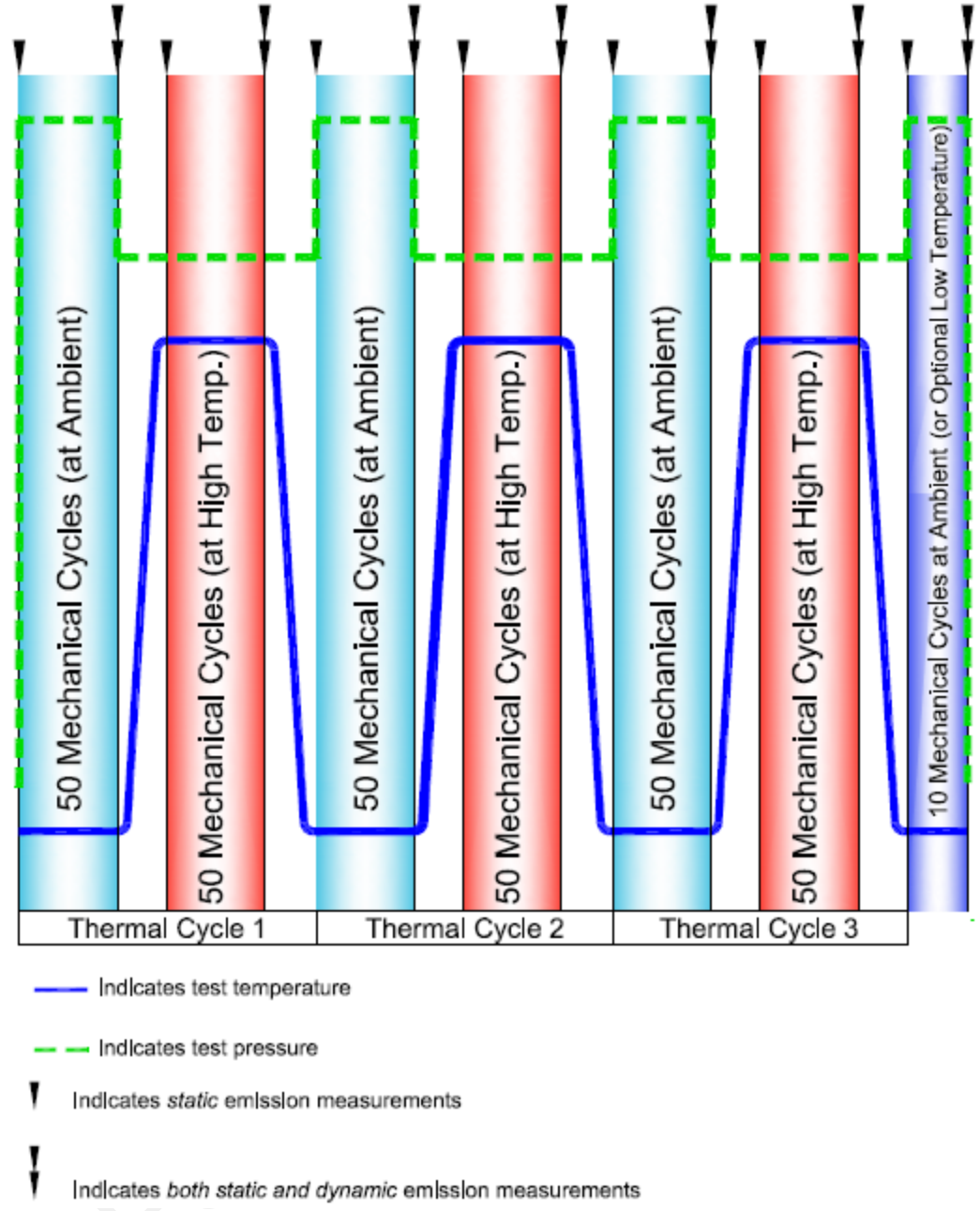


Figure 1 – Valve Cycling

5.2. The valve shall be heated using an external heat source such as electric heating blanket, coils, or other suitable equipment.

5.3. Test Temperature

The elevated test temperature shall be 260°C (500°F) +/-5 percent.

5.4. Test Pressure

The test pressure shall be in accordance with the temperature rating for the test valve per ASME B16.34 for the applicable body material and pressure class or 600 psig, whichever is lower.

5.5. The test valve may be equipped with a method of actuation capable of stroking the stem to simulate the mechanical cycle of a valve at the rate of 1 mm to 5 mm (0.04 in. to 0.20 in.) per second. The method of actuation should not impose additional side loads on the stem or packing. Running torque values shall be recorded on the first and last cycle of testing.

5.6. Leak Measurement

5.6.1. Packing leakage measurements around the full circumference of the stem OD and packing OD and the highest reading shall be recorded. Static and dynamic stem leakage measurements shall be taken. See Figure 1.

5.6.2. The static/dynamic leakage shall be recorded after a time delay of twice the response time of the instrument.

5.6.3. Leak measurements shall be sniffed using a detection probe.

5.6.4. Leakage from body-bonnet and auxiliary connections can affect leak measurement readings and shall be corrected prior to continuation of the test and noted in the report.

5.7. Temperature Monitoring

The valve shall be equipped with thermocouples for continuously monitoring the temperature during thermal cycling. The temperature shall be monitored and recorded at two locations (see Figure 2).

5.7.1. TC-1 - affixed to the stuffing box. The thermocouple affixed to the stuffing box shall control the test temperature.

5.7.2. TC-2 - at the external body adjacent to the flow path shall be a reference measurement.

5.7.3. The difference between TC-1 and TC-2 after stabilization shall be equal to or less than 10°C (50°F).

5.8. Packing adjustment during type testing is not permitted.

6. Acceptance Criteria

The Fugitive Emissions Test Report in Appendix A shall indicate “pass” when the measured leakage throughout the test does not exceed 100 ppmv.

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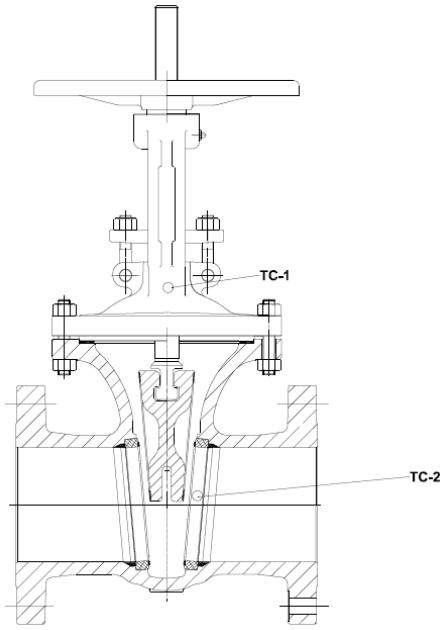


Figure 2 – Temperature Locations

7. Leak Test Equipment and Calibration

7.1. Monitoring equipment shall be as described below capable of providing on-board data logging with digital readout. The equipment shall be certified as intrinsically safe for use with the test medium.

7.2. The equipment shall meet the following performance requirements in the flame ionization mode:

a. Maximum variation: ± 2% at 100 ppmv.

b. Dynamic range: 1.0 – 50,000 ppmv.

c. Linear range: 1.0 – 10,000 ppmv.

d. Minimum detectable level (defined as 2 x the peak noise): 300 ppb hexane.

e. Maximum response time to reach final value: 3 seconds.

f. Maximum recovery time to return to 10% of initial value: 5 seconds.

g. Sample flow rate at probe inlet: 1.14 – 1.51 l/min (0.30 – 0.40 gal/min).

7.3. The testing equipment shall be calibrated daily prior to testing according to the manufacturer's directions. A record of test equipment calibration shall be maintained by the test facility.

7.4. The test equipment shall be inspected prior to each use to ensure against fouling of the detector probe. This shall be done per EPA Method 21 using an external calibration gas with a known methane concentration.

7.5. The emissions leak test system shall conform to all local and governmental safety standards and shall be equipped with pressure relief valve(s) / rupture disc(s) and vents as required.

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8. Post-test Inspection

8.1. Valve shall be disassembled and components including stem, packing, stem nut, gland follower, and stuffing box shall be inspected and the condition documented.

8.2. Sealing components shall be photographed in accordance with section 9.4.

9. Recording and Documentation

9.1. As a minimum, test results shall be provided on the Fugitive Emissions Test Report provided in Appendix A.

9.2. Leak measurements shall be recorded at established intervals as detailed in section 5.6. The testing facility shall issue a test certificate stating that the valve was tested in accordance with API 624.

9.3. Leakage rate units of measurement for methane is [ppmv].

9.4. Photographic evidence shall be documented as follows:

9.4.1. Valve marking and tagging.

9.4.2. Pre- and post-test valve external stuffing box area.

9.4.3. Post-disassembly valve components.

a. Stem packing area

b. Stem threads

c. Stem nut

d. Stem bushing

e. Packing gland follower

f. Stuffing box interior

g. Packing rings and spacers

10. Valves Qualified

10.1. When a valve has been successfully tested, it shall qualify the same type (i.e. rising or rising rotating). Other sizes may be tested subject to agreement between purchaser and manufacturer.

10.2. Pressure class qualification groupings shall be as defined below:

API 600 Valves

4”, 6”, & 12”	Class 300 qualifies classes 150, 300, and 600
4”	Class 1500 qualifies classes 900 and 1500
4”	Class 2500 qualifies only class 2500

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¾”	Class 800 qualifies classes 150, 300, 600, and 800
1”	Class 2500 qualifies classes 900, 1500, and 2500

10.3. Changes to the following items shall void the qualification of valves for which this standard was intended to qualify:

a) packing manufacturer, model, or type; or

b) design change that will reduce the strength or impair the operability or performance of the sealing system of the valve.

10.4 If the location of the valve manufacturing facilities is different than what is listed on the API 624 certificate, the purchaser may request re-qualification.

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ANNEX A - FUGITIVE EMISSIONS TEST REPORT

<table>
<tr><td colspan="3">API Std 624 Fugitive Emissions Testing Report Number: _____</td></tr>
<tr><td colspan="2">Valve Stem Type (check one):

( ) Rising ( ) Rotating-Rising</td><td>Manufacturer:

Valve Size, Class, & Type:

API/ASME Design Standard(s):</td></tr>
<tr><td>Packing Description:

Manufacturer:

Model/Type:

Gland follower exposed length:</td><td>Number of rings:

OD:

ID:

Stack height:

Gland follower insertion depth:</td><td>Body-bonnet material:</td></tr>
<tr><td colspan="2">Testing Facility:

Technician:

Witness:

Start Date: Completion Date:</td><td>Valve Selection (check one): Date Selected:

( ) Manufacturer

( ) Distributor

( ) Purchaser

Selected by:</td></tr>
<tr><td colspan="2">Gland nut recommended torque:

Body-bonnet bolting recommended torque:</td><td>Body-bonnet connection:

Type: ( ) Bolted

( ) Welded

( ) Pressure Seal

( ) Other _____</td></tr>
<tr><td colspan="3">Notes:</td></tr>
</table>

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API 624 Testing Data						
Test Segment	Cycle	Temperature @ TC1	Temperature @ TC2	Static Leak Measurement	Dynamic Leak Measurement	Other Leak Measurement (specify)
Ambient Temperature 0 – 50 cycles P = _____	0					
	50					
Elevated Temperature 51 – 100 cycles P = _____ T = _____	51					
	100					
Ambient Temperature 101 – 150 cycles P = _____	101					
	150					
Elevated Temperature 151 – 200 cycles P = _____ T = _____	151					
	200					
Ambient Temperature 201 - 250 cycles P = _____	201					
	250					
Elevated Temperature 251 - 300 cycles P = _____ T = _____	251					
	300					
Ambient (or Low) Temperature 301 - 310 cycles P = _____ T = _____	301					
	310					

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Running Torque Value: First Cycle: _____ Last Cycle:_____

Stem Orientation: _____

Pass ( ) Fail ( )

Lab Representative Signature: _____ Date: _____

Witnessed By: _____ Date: _____